Project Overview

This project develops a machine learning-based prediction system using the Linear Regression algorithm. The dataset is collected, analyzed, and preprocessed to prepare it for model training. The trained model is evaluated, saved using the Pickle library, and integrated into a Flask web application. Users can enter the required input values through the web interface, and the system generates accurate prediction results in real time. This project demonstrates the complete machine learning workflow, from data processing to model deployment.

Overall Project Flow

Problem Identification
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Environment Setup
↓
Dataset Collection
↓
Dataset Understanding (EDA)
↓
Data Preprocessing
↓
Train-Test Split
↓
Model Training (Linear Regression)
↓
Model Evaluation
↓
Save Model (Pickle)
↓
Flask Web Application Development
↓

Testing
↓
Final Prediction & Deployment